

Homework 2

Exponentials and logarithms · about 60 minutes

Each question graded N0–E8

Answers · tutor's working, not an official mark scheme

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

- 1 Write $5^3 = 125$ in log form. [A]

$$\log_5 125 = 3$$

- 2 Mia says $\log(10 + 10) = \log 10 + \log 10 = 2$. Is she right? Give a reason. [A]

No: there is no law for the log of a sum. $\log 20 = 1.301$

- 3 Write x^{-2} without a negative index. [A]

$$\frac{1}{x^2}$$

- 4 Factorise $x^2 - 49$. [A]

$$(x - 7)(x + 7)$$

QUESTION ONE

- (a) Evaluate (i) $\log_2 128$ (ii) $\log_{25} 5$ [A]

$$(i) 2^7 = 128, \text{ so } 7 \quad (ii) 25^{\frac{1}{2}} = 5, \text{ so } \frac{1}{2}$$

- (b) Solve $\log_x 343 = 3$. [A]

$$x^3 = 343, \quad x = 7$$

- (c) Write $2 \log(3a) - \log\left(\frac{a}{4}\right)$ as a single logarithm in its simplest form. [M]

$$\log 9a^2 - \log \frac{a}{4} = \log\left(9a^2 \times \frac{4}{a}\right) = \log 36a$$

u: the 2 moved up as a power; r: single log, simplified

- (d) Solve $\log_4(x - 1) + \log_4(x + 5) = 2$. [M]

$$(x - 1)(x + 5) = 4^2 = 16, \quad x^2 + 4x - 21 = 0, \quad (x + 7)(x - 3) = 0$$

$$x = 3 \quad (x = -7 \text{ rejected: } \log_4(-8) \text{ does not exist})$$

- (e) $\log_2 x + \log_2 y = 3$ and $\log_2 x - \log_2 y = 1$. Find x and y , explaining your reasoning. [E]

$$xy = 2^3 = 8 \quad \text{and} \quad \frac{x}{y} = 2^1 = 2$$

Multiplying: $x^2 = 16$, so $x = 4$ (not -4 : $\log_2 x$ needs $x > 0$), and $y = 2$.

u: (a), (b) correct; (c) first step. r: (c) correct; (d) correct with -7 rejected; (e) both equations in power form.
t: (e) $x = 4, y = 2$ with the negative value rejected for a reason. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION TWO

- (a) Solve $3^x = 40$. [A]

$$x = \frac{\log 40}{\log 3} = 3.358$$

- (b) Solve $2^{x+3} = 100$. [A]

$$(x+3) \log 2 = \log 100, \quad x+3 = \frac{\log 100}{\log 2} = 6.644, \quad x = 3.644$$

- (c) Solve $25^x = 125^{x-2}$ exactly. [M]

$$5^{2x} = 5^{3(x-2)}, \quad 2x = 3x - 6, \quad x = 6$$

- (d) Sam solved $4^{2x+1} = 60$. **Sam's answer has 2 errors.** Find each one, fix it, and say why it loses marks. [M]

Sam's answer

$$2x + 1 \log 4 = \log 60$$

$$2x + 1 = \log 15 = 1.176$$

$$x = 0.088$$

1. the power comes down in a bracket: $(2x+1) \log 4 = \log 60$ (L13)

2. $\log 60 \div \log 4$ is not $\log 15$: it is 2.9534, so $x = 0.977$ (L12)

- (e) Solve $9^x - 10 \times 3^x + 9 = 0$. [E]

$$\text{Let } u = 3^x, \text{ so } 9^x = u^2: \quad u^2 - 10u + 9 = 0, \quad (u-1)(u-9) = 0$$

$$3^x = 1 \text{ gives } x = 0; \quad 3^x = 9 \text{ gives } x = 2$$

u: (a), (b) correct. r: (c) exact answer; (d) both errors fixed with reasons; (e) the quadratic in u formed.
t: (e) both values of x found, with u changed back. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION THREE

A town of 8000 people grows by 3% each year, so its population is $P = 8000(1.03)^t$ after t years.

- (a) Find the population after 5 years. [A]

$$8000(1.03)^5 = 9274 \text{ people}$$

- (b) When will the population first reach 12 000? [A]

$$1.03^t = 1.5, \quad t = \frac{\log 1.5}{\log 1.03} = 13.72$$

It first reaches 12 000 during the 14th year.

- (c) A 400 mg dose of a medicine decays so that 250 mg remain after 6 hours. Find the percentage lost each hour. [M]

$$400r^6 = 250, \quad r = 0.625^{\frac{1}{6}} = 0.9247$$

About 7.5% of the medicine is lost each hour.

- (d) A kettle's water cools as $T = 22 + 78e^{-0.05t}$ (t in minutes). When is it 40°C ? [M]

$$78e^{-0.05t} = 18, \quad -0.05t = \ln \frac{18}{78}, \quad t = 29.33$$

The water reaches 40°C after about 29.3 minutes.

- (e) Account A holds \$5000 growing at 4% a year; account B holds \$4000 growing at 6% a year. After how many whole years does B first hold more than A? Show your reasoning with logs. [E]

$$4000(1.06)^t > 5000(1.04)^t \implies \left(\frac{1.06}{1.04}\right)^t > 1.25$$

$$t > \frac{\log 1.25}{\log \frac{1.06}{1.04}} = 11.71$$

B first holds more than A after 12 whole years.

u: (a), (b) correct. r: (c) rate found; (d) time found; (e) inequality set up.

t: (e) 12 years with the reasoning in context. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

Totals out of 24. Cut scores move a little each year: Achievement from about 7–8, Merit 13–14, Excellence 19–20.